Hima Shanthi Rayapati

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Education

Arizona State University, Data Science, Analytics and Engineering (Human-Centered Applications) 08/2026 – 12/2027

- Fulton Schools of Engineering Graduate Scholarship | ASU AI Scholars Program, 2026–27

Arizona State University, BS in Psychology and Mathematics (Statistics) 01/2023 – 05/2026

- **Minor:** Philosophy | **GPA:** 4.0/4.0

Research Experience

Research Assistant, *CARE Lab (Prof. Qiaoning Zhang)*, ASU 08/2026 – Present

- Attend lab meetings and discuss ongoing research on human-AI interaction.
- Developing my own research project on AI and human trust and learning (in progress).

UX Researcher (UXR Guild), *NEXT Lab*, ASU 08/2026 – Present

- Test games, digital twins, and interactive apps with users to find usability problems and collect feedback.
- Find and document bugs and performance issues and share them with developers.
- Work with developers on UI and accessibility improvements.

Research Assistant, *Thinking Across Languages and Contexts Lab*, ASU 08/2025 – Present

- Assist a graduate student with testing experiments built in jsPsych and JavaScript.
- Help check whether data are recorded correctly and organized for analysis.
- Perform exploratory data analysis and assist in interpreting experimental results under supervision.

Research Assistant, *Perception, Action, Language & Meaning Lab*, ASU 08/2025 – Present

- Run two-person collaborative task sessions (e.g., Taboo), observing and recording coordination, communication, and performance data using VR, motion-capture, and eye-tracking.
- Work with the lab's time-series behavioral data, learning how to clean, organize, and explore it.
- Read papers and explore possible methods (e.g., multilevel vector autoregression) to understand which approaches might fit the lab's data.
- Contribute to discussions by sharing preliminary notes and observations from trial analyses.

Volunteer Research Assistant, *Longitudinal & Behavioral Data Science Lab*, ASU 08/2024 – 12/2024

- Participated in bi-weekly lab meetings, contributing to discussions and staying updated on research progress.

Research Assistant, *Evolution, Ecology, and Social Behavior Lab*, ASU 08/2023 – 11/2023

- Assisted with survey design, pilot testing, data coding, and literature reviews in social cognition research.
- Participated in lab meetings and contributed to discussions on study design and interpretation.

Projects

Trust in Advice: AI vs. Human Sources (Data Analysis & Written Report) ***Trust in AI.com/2025***

Tools: SAS (PROC GLMSELECT, PROC GLM)
Methods: General linear model, Stepwise selection (AIC), Interaction effects, Residual diagnostics, Correlation analysis

- Analyzed 30,000+ human-AI interactions to study effects of advice source.
- Modeled decision changes before and after advice exposure.
- Found advice correctness as primary driver, with modest AI-related effects.

Analyzing the Impact of AI on Stack Overflow Participation (NLP & Data Analysis) ***GitHub***

Tools: Python (pandas, NumPy, Matplotlib, Seaborn, statsmodels, SciPy), SQL, Jupyter, Google Colab
Methods: Time-series analysis, Statistical comparisons, Engagement metrics, NLP

- Analyzed large-scale Stack Overflow data using Python to study changes in user behavior after AI tool releases.
- Built time-series datasets and computed engagement metrics (answer rates, response time, retention) using

Python.

- Performed statistical comparisons and visualized platform-level trends.
- Explored tag-based trends and basic NLP features to examine topic shifts.

Cultural Differences in Trust: AI vs. Human Expert Decision-Making ***Research Proposal / 2025***

Methods: Mixed-effects modeling, Experimental design, Pre-registration, Literature synthesis

- Proposed cross-cultural study (Mexico vs. U.S.) on AI vs. human decision-making under conflict.
- Outlined experimental design focusing on competence and legitimacy.
- Formulated mixed-effects modeling approach for analysis.
- Synthesized literature across cognition, decision-making, and human-AI interaction.

Fairness in ML: Detecting Bias in LLM Answers (CSE 575, in progress)

Methods: Text classification, Bias evaluation (BBQ benchmark), LLM answer selection

- Building a classifier that flags biased answers from an LLM, using the BBQ bias benchmark dataset.
- Testing whether the classifier can pick the least biased answer out of several options the LLM gives.

Stochastic Dual-Timescale Model of Learning (ODE & SDE Modeling) ***Final Project / 2025***

Tools: MATLAB, R Shiny

Methods: ODE modeling, SDE modeling, Stability analysis, Simulation

- Developed differential equation model of fast and slow learning processes.
- Extended to stochastic framework to examine effects of noise.
- Verified theoretical properties via simulation.
- Linked model behavior to memory retention and learning variability.

Probabilistic Reward Learning Task using PsychoPy ***Reward Learning.com/2025***

Tools: PsychoPy, Python (pandas, NumPy)

Methods: Experimental design, Learning under uncertainty, Behavioral data analysis

- Designed probabilistic reward-learning task with informative vs. non-informative features.
- Examined learning under uncertainty and perceptual noise.
- Collected and analyzed participant data using Python.

Predictive Analysis of Team Performance in Baseball using SAS ***Baseball Analysis.com/2024***

Tools: SAS

Methods: Multiple regression, EDA, Correlation analysis, Multicollinearity checks, Model assumption validation

- Developed a statistical model to predict team wins using historical baseball data.
- Conducted EDA and correlation analysis on Minnesota Twins batting records (1998-2024, excluding 2020).
- Built multiple regression models to identify key predictors of team wins, selecting runs and batting averages for a parsimonious model.
- Addressed multicollinearity and validated model assumptions to improve predictive accuracy.

Predicting Daily Stock Prices of ADP using SAS ***ADP Stock.com/2024***

Tools: SAS

Methods: Multiple regression (stepwise, forward, backward), Quadratic regression, EDA, Correlation analysis

- Created a predictive model for ADP stock prices based on financial indicators.
- Applied EDA, correlation analysis, and multiple regression (stepwise, backward, forward selection) to determine significant predictors.
- Developed a quadratic regression model with strong explanatory power while addressing multicollinearity.
- Examined sector-specific dynamics and market trends influencing ADP's stock behavior.

Cyber Attack Detection in Water System Sensor Data (CSE 575, in progress)

Methods: Anomaly detection, Time-series sensor data

- Group project using the BATADAL water distribution system dataset to detect cyber attacks from sensor readings.
- Helping with the project proposal, report, and presentation.

Research Proposal: Mirror Neurons, Social Synchrony, and Collaborative Learning ***Proposal.com/2025***

- Proposed a research framework synthesizing human and animal literature on mirror neuron systems and social synchrony.
- Formulated hypotheses linking coordination mechanisms to collaborative learning outcomes.

Additional projects and full write-ups available at my portfolio.

Work Experience

Graduate Course Assistant (Statistical Methods), ASU – Tempe, AZ 01/2027 – Present

- Grade assignments and exams and give feedback to students.
- Hold office hours and answer student questions about course content.
- Work under the direction of a faculty supervisor.

Research Aide, ASU – Tempe, AZ 12/2024 – Present

- Review, categorize, and structure user-generated data to ensure quality and consistency.
- Assign relevant tags to comments for improved organization and searchability.
- Evaluate AI-generated tags against human annotation standards, analyzing human-AI agreement and error patterns across prompt versions to guide model refinement.
- Perform data cleaning tasks, identifying and correcting inconsistencies to maintain data accuracy.
- Work with databases to structure data and ensure accessibility for future use.

Lead Survey Assistant, ASU – Tempe, AZ 05/2023 – Present

- Conduct phone-based survey interviews with currently employed and recently graduated ASU students, ensuring accurate and comprehensive data collection by clarifying responses and addressing any ambiguities.
- Assist in the hiring process, including interviewing and evaluating candidates.
- Perform data cleaning and ensure the integrity of datasets for analysis.
- Train new student workers in survey procedures and oversee testing and calibration of survey tools.

Intern, Public Safety Crisis Solution – Phoenix, AZ 08/2024 – 11/2024

- Performed quantitative analysis to support research, program development, and accreditation documentation.
- Contributed research-based content on clinical procedures addressing first responders' mental health initiatives.

Advising Office Aide, ASU – Tempe, AZ 08/2023 – 05/2024

- Supported advising operations through scheduling, communication, and administrative workflows.
- Managed data entry, marketing/outreach materials, and front-desk support for students and staff.

Skills

- **Programming & Statistical Tools:** R, SAS, MATLAB, Python, JMP, JASP, SQL (Microsoft Access), Java, C, Bash, REST APIs
- **Machine Learning & NLP (Foundational):** Principal Component Analysis (PCA), Neural network basics, Supervised vs. Unsupervised learning, Model evaluation concepts, Introductory text analysis (Voyant Tools)
- **Data Analysis & Visualization:** Tableau, Excel, Google Sheets, Alteryx
- **Experimental & Research Tools:** jsPsych, PsychoPy, JavaScript, HTML, CSS, Qualtrics, SurveyMonkey
- **Design & Prototyping:** Figma, Figma Motion
- **Collaboration & Productivity:** Slack, Teams, Zoom, Word, Outlook, Adobe PowerPoint, OneDrive, Dropbox